Power Generation Methods

DAY-1 [8/9/25]

* Cost of electrical energy per kWhr must be minimum
    - Power Generation Methods
    - Power Transmission methods
    - Economic factors
    - Economic load Dispatch
    - Load flow studies
    - HVDC

* Rated Vtg has to be available within the safe limits
    -> Overvtg causes damage to the insulation.
    -> Undervtg causes overheating in the equipment due to high current.

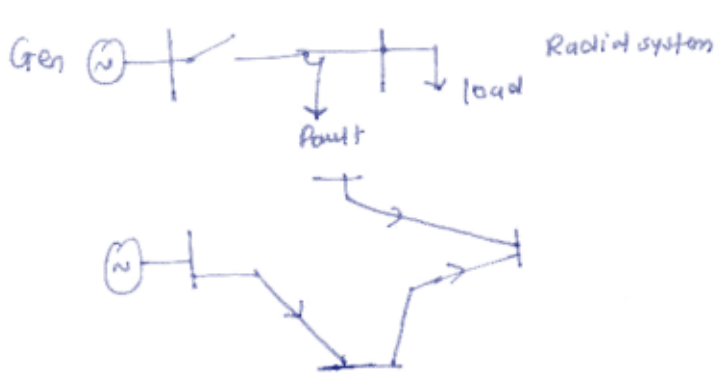
* Rated frequency of 50 Hz has to be maintained within the limits
    -> Load frequency control :
    At low frequency X'mer core is saturated and performance of motors are affected.
    [$\because V = 4.44 f \Phi_m N$]

* At high frequency hysteresis losses are increased.

* Faster fault identification and isolation of fault from the healthy system has to be available - Fault analysis, switchgear & protection.

* More stable generator has to be available w/o losing synchronism during temporary fault conditions - "Stability".

* Continuous reliable power supply has to be available to all the consumers - "Transmission, distribution, HVDC".

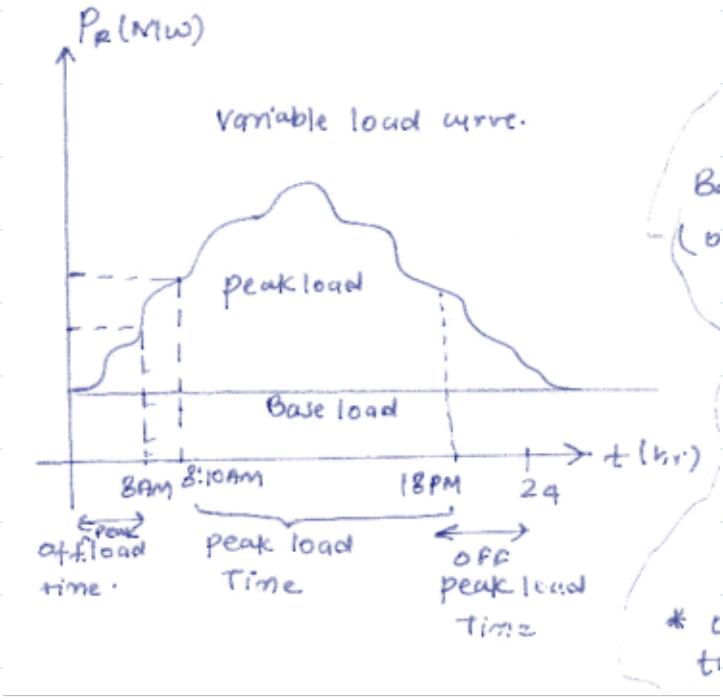


Radial system / interconnected system

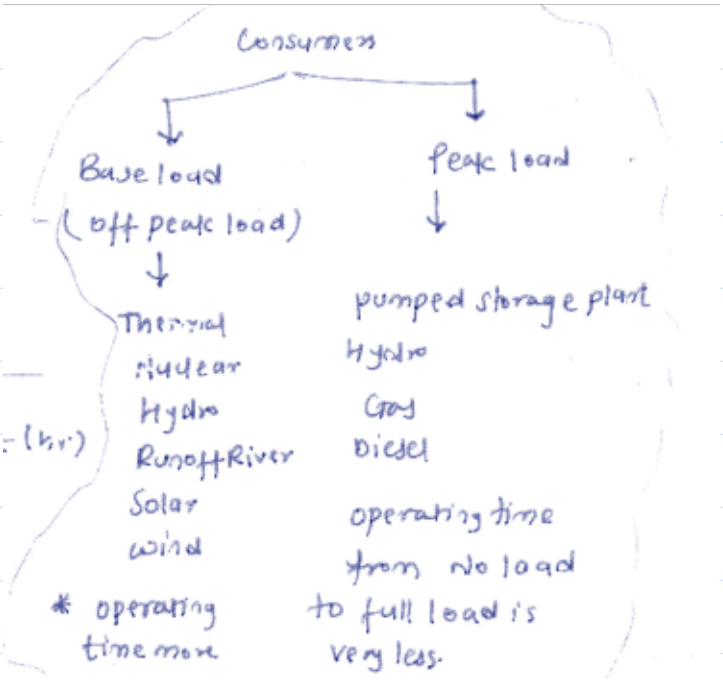
* Flexible power transmission has to be available in all vtg levels - 'power cable'.

* Per capita power consumption in India is around 1,500 units.

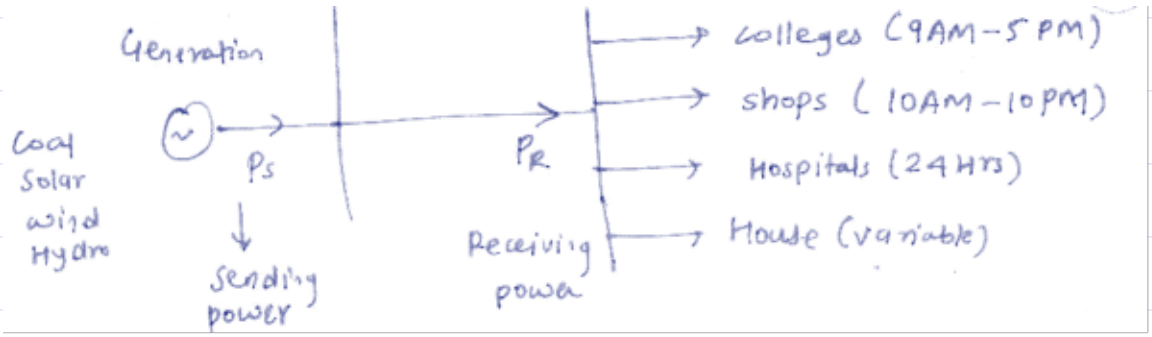
* Total generator installed capacity in India is 500 GW, which is obtained from coal or thermal plants 47.63%, renewable energy 32.5%, hydro 10.6%, gas 5.6%, nuclear 1.85%, diesel 0.13%.



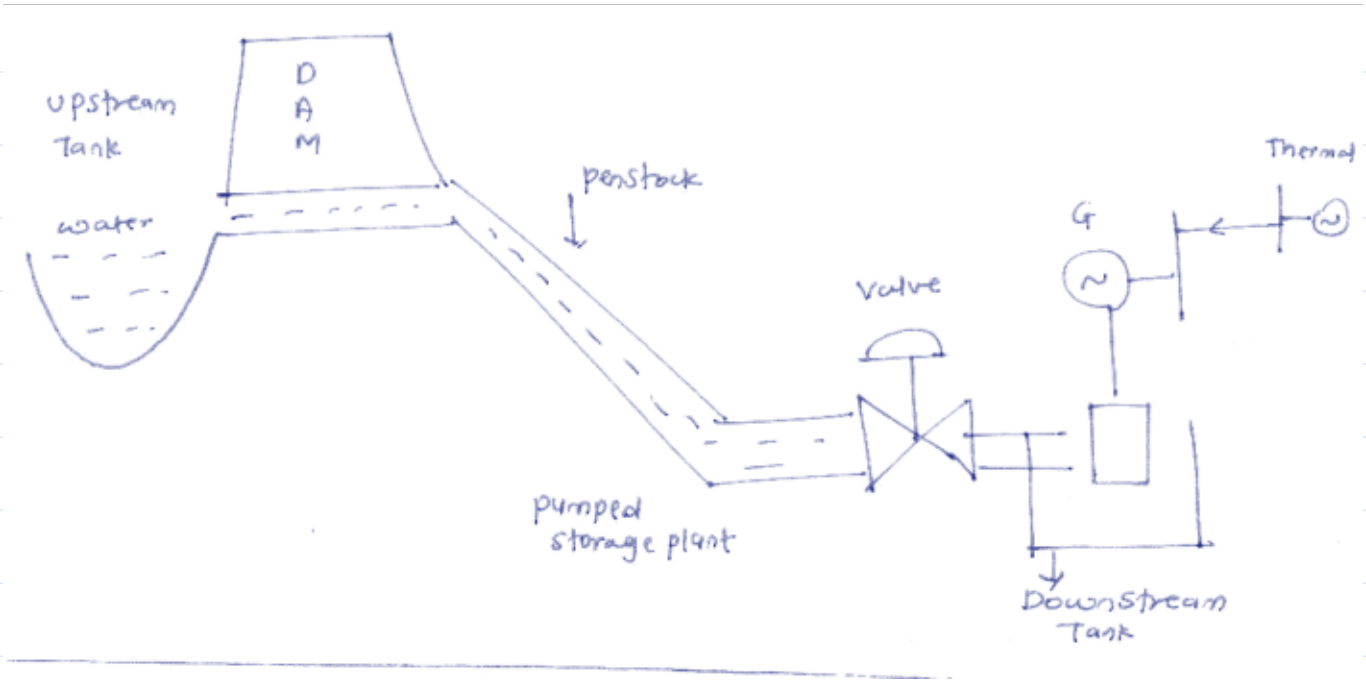
Variable load curve: peak load and base load



Consumers: base load and peak load plants



Generation, sending power Ps, receiving power PR and loads



Pumped storage plant

* Out of 500 GW of total power generation, 35%, 28%, 25%, 12%, 1% are obtained from West, South, North, East, North-Eastern region respectively.

* Energy consumption in India: Domestic (30%), Industrial (28%), Irrigation (21%), Commercial (9%), Others - Railways (12%).